



Contents lists available at ScienceDirect

Journal of Complexity

journal homepage: www.elsevier.com/locate/jco

High-dimensional quasi-Monte Carlo via combinatorial discrepancy

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ARTICLE INFO

Article history:

Received 4 September 2025

Received in revised form 30 April 2026

Accepted 4 May 2026

Available online 5 May 2026

Communicated by C. Aistleitner

MSC:

primary 65D30

secondary 11K38, 11K45

Keywords:

Quasi-Monte Carlo

High-dimensional integration

Combinatorial discrepancy

Numerical integration

ABSTRACT

Quasi-Monte Carlo (QMC) methods are known to achieve faster convergence rates than Monte Carlo (MC), but their effectiveness in high dimensions often relies on additional structure, such as low effective dimension or carefully chosen coordinate weights. Moreover, in many applications one has access only to random samples rather than deterministic QMC constructions. In this work, we extend the recent method of N. Bansal and H. Jiang and construct high-dimensional QMC point sets from random samples via combinatorial discrepancy. We establish error bounds for these constructions in weighted function spaces, including settings with low effective dimension in both the superposition and truncation senses. We also implement the resulting Subgaussian Transference algorithm and present numerical experiments to assess empirically the performance of these constructions.

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1. Introduction

In this work, we will study the problem of approximating

$$\mu := \int_{[0,1]^d} f(\mathbf{x}) d\mathbf{x},$$

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<https://doi.org/10.1016/j.jco.2026.102053>

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where the dimension d is often large. *Quasi-Monte Carlo (QMC) methods* [12,7,20] can be implemented in this setting in the hope to improve upon the naive Monte Carlo (MC) method by replacing the random samples in the sample mean approximation, $\hat{\mu}$, with low-discrepancy point sets and sequences. The convergence rates of QMC methods are captured by one of several variants of the Koksma-Hlawka inequalities [15,13], which bounds the integration error as

$$|\hat{\mu}_n - \mu| \leq D(\{\mathbf{x}_i\}_{i=1}^n) \cdot V_{\text{HK}}(f), \quad (1)$$

where $V_{\text{HK}}(f)$ is the variation of f in the sense of Hardy and Krause [23], and $D(\cdot)$ defined as

$$D(\{\mathbf{x}_i\}_{i=1}^n) := \sup_{\mathbf{a} \in [0,1]^d} \left| \frac{1}{n} \sum_{i=1}^n \mathbf{1}(\mathbf{x}_i \in [\mathbf{0}, \mathbf{a}]) - \text{vol}([\mathbf{0}, \mathbf{a}]) \right| \quad (2)$$

measures the uniformity of the sampling nodes with respect to axis-aligned origin-anchored test boxes, and is referred to as the *star discrepancy*.

For a given low-discrepancy point set $\{\mathbf{x}_i\}_{i=1}^n$, one can bound its star discrepancy by $C(\log n)^{d-1}/n$ for some constant C depending on the specific construction. This implies a deterministic integration error of the same order whenever $V_{\text{HK}}(f)$ is finite. Compared with the MC error rate $O(n^{-1/2})$, the dependence on n is asymptotically superior provided the dimension d is not too large.

1.1. Beyond Hardy-Krause via combinatorial discrepancy

In practice, the variation $V_{\text{HK}}(f)$ can be orders of magnitude larger than $\sigma(f)$, especially for functions with oscillatory or high-frequency components. Worse, the classical Koksma-Hlawka (1) bound ties QMC performance to $V_{\text{HK}}(f)$, which is typically infinite for discontinuous integrands, so even implementing a low-discrepancy sampling node may yield a pessimistic bound. Furthermore, classical QMC—in the sense of requiring a deterministic node construction—cannot be readily applied when only random samples are available.

To address these issues, recent work by N. Bansal and H. Jiang [3] gives a randomized QMC construction that offers significantly tighter error bounds—they replace $V_{\text{HK}}(f)$ in (1) with a “smoothed-out variation”, $\sigma_{\text{SO}}(f)$, which is substantially smaller for many practical integrands, especially those with substantial high-frequency components. Their QMC method is based on the notion of *combinatorial discrepancy*, that quantifies the necessary imbalance in any ± 1 coloring of a finite point set with respect to a family of subsets $\mathcal{F} \subseteq 2^{[n]}$, and is formally defined as

$$\text{disc}(\{\mathbf{x}_i\}_{i=1}^n, \mathcal{F}) := \min_{\chi: \{\mathbf{x}_i\}_{i=1}^n \rightarrow \{\pm 1\}} \max_{F \in \mathcal{F}} \left| \sum_{i \in F} \chi(\mathbf{x}_i) \right|.$$

Heuristically, combinatorial discrepancy measures how evenly a finite point set can be split into two halves across a prescribed family of test sets. Indeed, if one colors the points by signs ± 1 , then a small value of

$$\max_{F \in \mathcal{F}} \left| \sum_{i \in F} \chi(\mathbf{x}_i) \right|$$

means that, for every test set $F \in \mathcal{F}$, the numbers of positively and negatively colored points inside F are nearly the same. In this sense, low combinatorial discrepancy provides a way to partition a point set into two subsets that remain similarly distributed with respect to the family $\mathcal{F}$. Repeating such nearly balanced splittings is the basic mechanism behind the QMC constructions studied in [3].

Despite the different-looking definition, combinatorial discrepancy is known to be closely connected to star discrepancy through the *transference principle*; see [21,1], which, in the QMC context, states that low-discrepancy point sets can be obtained from colorings with small combinatorial discrepancy with respect to anchored boxes (see Lemma 4, Appendix A.1.1). Leveraging the latest algorithmic developments in combinatorial discrepancy theory and the transference principle, [3] obtained their novel QMC construction with the improved error bounds.

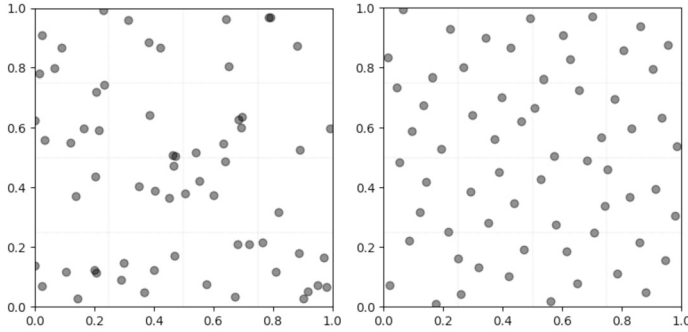


Fig. 1. 64-element point sets in $[0, 1]^2$. (Left) Uniform IID points (Right) Points A_T resulting from algorithm SubgTrans with a Sobol' initialization.

1.2. Quasi-Monte Carlo in high dimension

The discussion so far has assumed fixed dimension d and sufficiently large n , where QMC is known to outperform MC in convergence rate. In high dimensions, however, classical discrepancy bounds of order $(\log n)^d/n$ deteriorate rapidly with d , and can become substantially worse than the MC rate $O(n^{-1/2})$; see [12, Section 5.3]. Taken at face value, this appears to undermine the usefulness of QMC for high-dimensional integration, yet, this pessimistic conclusion is often contradicted in practice. To explain the observed empirical success of QMC in high-dimensional problems, the notion of *effective dimension* emerged [26,4,27], reflecting the fact that many integrands depend strongly only on a small subset of variables or on low-order interactions. (Interestingly, the first paper to mention “quasi-Monte Carlo” [28] considered a precursor to this idea [25].) These ideas were later formalized through two notions of effective dimension in the *truncation* and *superposition* sense (see Section 3 for formal definitions).

A complementary rigorous framework arose through weighted function spaces [31], in which variable importance is incorporated directly into the analysis. In this setting, one often assigns coordinate weights $\gamma_j \geq 0$ for $1 \leq j \leq d$, typically satisfying $\gamma_1 \geq \gamma_2 \geq \dots \geq \gamma_d \geq 0$, and for subsets $u \subseteq [d] := \{1, 2, \dots, d\}$ defines product weights $\gamma_u := \prod_{j \in u} \gamma_j$. Under suitable decay conditions on the weights, tractability theory shows that QMC can achieve dimension-independent convergence rates [29]. This framework has led to a large literature on high-dimensional QMC including the popular component-by-component construction [16,30,22,18]. On the other hand, it is worth mentioning that several recent works circumvent the difficulty of choosing weights by proposing ‘universal’ algorithms based on median-of-means estimators [11,10].

In any case, all approaches proposed above assume that one is free to design the sampling nodes. In many modern applications, one instead has access only to a population of random samples. This raises the motivating question for our work: can one transform such random samples into point sets with QMC-type error guarantees in high dimensions?

1.3. Our contributions

Following the work of Bansal and Jiang [3], we study new constructions of high dimensional QMC sets via combinatorial discrepancy theory. In particular, we adapt their constructions for integrands belonging to weighted function spaces and for functions with low effective dimension.

For a fixed choice of product weights $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \dots, \gamma_d)$, the algorithm described in Section 2 constructs a QMC point set $A_T = A_T(\boldsymbol{\gamma})$; see Fig. 1. We prove the following error bound for weighted function spaces.

Theorem 1 (Weighted subgaussian transference). *For every square-integrable function $f : [0, 1]^d \rightarrow \mathbb{R}$ with finite weighted smoothed-out variation as defined in (3), the integration error satisfies*

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O\left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n^2} \cdot \sigma_{\text{WSO}}(f)^2\right),$$

where the weighted smoothed-out variation

$$\sigma_{\text{WSO}}(f)^2 := \sum_{\mathbf{k} \neq \mathbf{0}} |\hat{f}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} \frac{|k_j|}{\gamma_j^2} \right). \quad (3)$$

We can obtain similar error bounds for functions with low effective dimensions, but we defer the formal statements to Section 3. We also remark that as in [3], the error rate of these QMC constructions is at most that of the MC error (up to a logarithmic factor), thereby achieving the best of both the error in Section 1 and the MC error.

Numerical Experiments. We also implement the subgaussian transference algorithm, providing the first numerical evaluations of the QMC constructions in [3] and the weighted extensions introduced here. Our experiments evaluate the irregularity of distribution of the resulting point sets and integration tests in both truncation and superposition dimension settings. The results investigate the robustness of subgaussian transference across different initializations, its consistent improvement over using those random samples for MC integration.

Roadmap. The structure of this paper is as follows. Section 2 introduces the weighted subgaussian transference algorithm, WSubgTrans, and establishes its theoretical properties. Section 3 develops variants of the subgaussian transference algorithm tailored to functions with low effective dimension and analyzes their theoretical guarantees. The proofs of the main results for the weighted setting (Section 2) and the effective-dimension variants (Section 3) are provided in Appendix A.1 and Appendix A.2, respectively. Section 4 reports numerical experiments considering both irregularity of distribution and high-dimensional integration approximation examples. Finally, Section 5 discusses the implications of our findings and outlines directions for future work.

2. The QMC construction for weighted spaces

In this section, we present a variant of the subgaussian transference algorithm from [3], adapted to weighted function spaces, and analyze its theoretical properties. We refer to this algorithm as WSubgTrans throughout. Subsection 2.1 introduces the necessary notation, and Subsection 2.2 describes the algorithm in detail.

2.1. Notation

For integer $d > 0$, we write $[d] := \{1, \dots, d\}$. We use bold lowercase letters, e.g., $\mathbf{z} = (z_1, \dots, z_d)$, to denote vectors, with z_j denoting the j^{th} coordinate of $\mathbf{z}$. For a collection of vectors, we denote the i^{th} vector by $\mathbf{z}_i$, and refer to its k^{th} coordinate as $z_{i,k}$. Given a function $f : [0, 1]^d \rightarrow \mathbb{R}$ and a finite set of points $A \subseteq [0, 1]^d$, we define the (signed) integration error with respect to point set A .

$$\text{err}(A, f) := \frac{1}{|A|} \sum_{\mathbf{x} \in A} f(\mathbf{x}) - \int_{[0, 1]^d} f(\mathbf{x}) d\mathbf{x}.$$

Dyadic Intervals and Boxes. Let $j \in \mathbb{N}$, we denote by $\mathcal{D}_j$ the set of all left-open dyadic intervals in $(0, 1]$ of length 2^{-j} , i.e., $\mathcal{D}_j := \{(\ell/2^j, (\ell+1)/2^j] : \ell = 0, \dots, 2^j - 1\}$, and call j the level of these dyadic intervals. For $h \in \mathbb{N}$, define $\mathcal{D}_{\leq h} := \bigcup_{j \leq h} \mathcal{D}_j$ as the set of all dyadic intervals of level less than or equal to h . In higher dimensions, $\mathcal{D}_{\leq h}^{\otimes d} := \{I_1 \times \dots \times I_d : I_j \in \mathcal{D}_{\leq h} \text{ for all } j \in [d]\}$ denotes all d -dimensional dyadic boxes with level at most h in each dimension.

Product Weights. Let $\gamma_1 \geq \gamma_2 \geq \dots \geq \gamma_d > 0$ be a sequence of positive weights assigned to each coordinate direction as in Subsection 1.2. For any dyadic box $B = B_1 \times \dots \times B_d \in \mathcal{D}_{\leq h}^{\otimes d}$, we say B is

trivial in dimension j if $B_j = (0, 1]$. The *weight* of B , denoted as $\gamma(B)$, is then defined as the product of the weights corresponding to its nontrivial coordinate directions:

$$\gamma(B) := \prod_{j \in [d]: B_j \neq (0, 1]} \gamma_j, \quad (4)$$

with the convention that $\prod_{j \in \emptyset} \gamma_j = 1$. For nonempty subset $u \subseteq [d]$, we slightly abuse notation and write $\gamma(u) := \prod_{j \in u} \gamma_j$. Lastly, throughout, we assume without loss of generality that n is an integer power of 2. Indeed, for arbitrary n , letting $m = \lfloor \log_2 n \rfloor$, we have $2^m \leq n < 2^{m+1}$. Thus 2^m and n differ by at most a factor of 2, so restricting to dyadic n incurs only an absolute constant factor in the relevant estimates.

2.2. WSubgTrans algorithm

The WSubgTrans algorithm aims to produce n point sets, each of size n , with small continuous discrepancy, while accounting for product weights across coordinate directions.

High-level overview. The construction starts from n^2 independent random samples and recursively halves the current point sets. At each stage, we assign a balanced $\{\pm 1\}$ -coloring to the points so that, across many weighted dyadic boxes simultaneously, the two color classes are as evenly distributed as possible. Splitting according to this coloring yields smaller subsets that still approximate the target volume well on those boxes. Repeating this procedure over $\log_2 n$ stages produces final point sets with small continuous discrepancy. In this way, combinatorial discrepancy controls the imbalance introduced at each halving step, while the transference principle converts these local imbalance bounds into the final discrepancy bound.

To make this precise, each point is encoded by a vector whose coordinates record its membership in the relevant weighted dyadic boxes. The resulting task is a vector balancing problem: we seek a balanced coloring that keeps the signed sum of these vectors small in ℓ_∞ , thereby ensuring that the split is nearly balanced across all relevant boxes at once.

More concretely, the vector balancing problem underlying this procedure is as follows: given vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^m$ of ℓ_2 -norm at most 1, and the goal is to find a coloring $\mathbf{c} = (c_1, \dots, c_n) \in \{\pm 1\}^n$ to minimize the combinatorial discrepancy $\|\sum_{j=1}^n c_j \mathbf{v}_j\|_\infty$. To address this, we first recall the Self-Balancing Walk algorithm from [2], which solves the problem in an online manner by sequentially assigning signs to the input vectors while ensuring small discrepancy with probability at least $1 - \delta$.

Algorithm 1 Self-Balancing Walk [2].

```

1:  $\mathbf{w}_0 \leftarrow 0$ 
2:  $\lambda \leftarrow 30 \log(mn/\delta)$ 
3: for  $1 \leq j \leq n$  do
4:   if  $|\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle| > \lambda$  or  $\|\mathbf{w}_{j-1}\|_\infty > \lambda$  then
5:     Fail ▷ Algorithm terminates with failure
6:   end if
7:    $p_j \leftarrow \frac{1}{2} - \frac{\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle}{2\lambda}$ 
8:    $c_j \leftarrow 1$  with probability  $p_j$ , and  $c_j \leftarrow -1$  with probability  $1 - p_j$ 
9:    $\mathbf{w}_j \leftarrow \mathbf{w}_{j-1} + c_j \mathbf{v}_j$ 
10: end for
11: return coloring  $\mathbf{c} = (c_1, \dots, c_n) \in \{\pm 1\}^n$ 

```

The idea behind Self-Balancing Walk is as follows. At step j , the algorithm maintains the partial sum $\mathbf{w}_{j-1}$ and decides the sign of $\mathbf{v}_j$ probabilistically depending on $\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle$. If $\mathbf{w}_{j-1}$ and $\mathbf{v}_j$ are orthogonal, it chooses $c_j = \pm 1$ with equal probability. As the alignment $\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle$ increases, the probability of selecting $c_j = -1$ increases, counteracting drift in that direction. The parameter $\lambda = 30 \log(mn/\delta)$ ensures that, with probability at least $1 - \delta$, $|\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle|$, $\|\mathbf{w}_{j-1}\|_\infty \leq \lambda$, so the iterations remain feasible with high probability. A formal performance guarantee is given in Theorem 6 (Appendix A.1).

To ensure that the coloring $\mathbf{c} = (c_1, \dots, c_n)$ is *balanced*, i.e., has equal number of 1's and -1 's, [3] proposed a modified algorithm BalSubgDisc. For even n , this is achieved by pairing adjacent vectors to form $\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3 - \mathbf{v}_4, \dots, \mathbf{v}_{n-1} - \mathbf{v}_n$, and then applying Self-Balancing Walk to the resulting set. We will assume the balanced property henceforth.

Our WSubgTrans algorithm, which constructs product-weighted low discrepancy point sets, is formally defined as follows:

Algorithm 2 WSubgTrans.

```

1:  $h \leftarrow O(\log(dn))$ 
2: Sample a random shift  $\mathbf{s} \sim [0, 1)^d$  and shift the dyadic system  $\mathcal{D}_{\leq h}^{\otimes d}$  accordingly (fold coordinates exceeding 1 back into  $[0, 1)$ ). ▷ For exposition, take  $\mathbf{s} = \mathbf{0}$ 
3:  $n_0 \leftarrow n^2, \quad T \leftarrow \log_2 n$ 
4:  $A_0^{(0)} \leftarrow n_0$  IID uniformly distributed samples from  $[0, 1]^d$ 
5: for  $t = 0$  to  $T - 1$  do
6:   for  $i = 0$  to  $2^t - 1$  do
7:      $A_t \leftarrow A_t^{(i)}$ 
8:     for each  $\mathbf{z}_j \in A_t$  do
9:        $\mathbf{v}_j \leftarrow (\mathbf{e}_j^{A_t}, (\gamma(B)\mathbf{1}_{\{\mathbf{z}_j \in B\}})_{B \in \mathcal{D}_{\leq h}^{\otimes d}})$  ▷  $\mathbf{e}_j^{A_t} \in \{0, 1\}^{|A_t|}$ , with a single one in the  $j$ th coordinate
10:    end for
11:     $\mathbf{c}_t \leftarrow \text{BalSubgDisc}(\{\mathbf{v}_j\})$  ▷  $\mathbf{c}_t \in \{\pm 1\}^{|A_t|}$ : balanced coloring
12:     $A_{t+1}^{(2i)} \leftarrow \{\mathbf{z}_j \in A_t : c_{t,j} = -1\}$ 
13:     $A_{t+1}^{(2i+1)} \leftarrow \{\mathbf{z}_j \in A_t : c_{t,j} = +1\}$ 
14:  end for
15: end for
16: return  $\{A_T^{(i)} : 0 \leq i < 2^T\}$  ▷  $2^T (= n)$  final sets, each of size  $n$ 

```

For the reader's convenience, we repeat Theorem 1 here. It provides the theoretical guarantee for the point sets A_T generated by the WSubgTrans algorithm, and its proof is given in Appendix A.1.

Theorem 1 (Weighted subgaussian transference). *For every square-integrable function $f : [0, 1]^d \rightarrow \mathbb{R}$ with finite weighted smoothed-out variation as defined in (3), the integration error satisfies*

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O\left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n^2} \cdot \sigma_{\text{WSO}}(f)^2\right),$$

where the weighted smoothed-out variation

$$\sigma_{\text{WSO}}(f)^2 := \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} \frac{|k_j|}{\gamma_j^2} \right). \quad (3)$$

Remark 1. The WSubgTrans algorithm depends on the weights $\gamma_1 \geq \gamma_2 \geq \dots \geq \gamma_d > 0$ assigned to coordinate directions. In the unweighted case ($\gamma_j \equiv 1$), it reduces to the subgaussian transference algorithm of [3], hereafter denoted SubgTrans. Under this specialization, Theorem 1 recovers [3, Theorem 1.1], yielding the bound $\frac{(\log n)^{O(d)}}{n^2} \sigma_{\text{SO}}(f)^2$, where $\sigma_{\text{SO}}(f)^2$ (“smoothed-out variation”) coincides with $\sigma_{\text{WSO}}(f)^2$ when $\gamma_j \equiv 1$. This bound becomes vacuous in high dimensions due to the $(\log n)^{O(d)}$ factor. By contrast, for product weights $\gamma_j = j^{-\alpha}$ with some $\alpha > 1/2$,

$$\prod_{j \in [d]} (1 + \gamma_j^2 \log n) = \prod_{j \in [d]} \left(1 + \frac{\log n}{j^{2\alpha}} \right) \leq \exp(C_\alpha (\log n)^{\frac{1}{2\alpha}}),$$

where $C_\alpha = 2\alpha + 1 + \frac{1}{2\alpha-1}$. In particular, the right-hand side is dimension-independent and subpolynomial in n : for any $\varepsilon > 0$, $\exp(C_\alpha (\log n)^{\frac{1}{2\alpha}}) = o(n^\varepsilon)$ as $n \rightarrow \infty$.

Remark 2 (Connection with the weighted Korobov space). The (periodic) weighted Korobov space of smoothness $\alpha > 0$ with product weights $\{\gamma_j\}$ is defined (see, e.g., [32,8]) by

$$\|f\|_{K_{\alpha,\gamma}}^2 = \sum_{\mathbf{k} \in \mathbb{Z}^d} |\widehat{f}(\mathbf{k})|^2 \prod_{j=1}^d \left(1 + \frac{|k_j|^{2\alpha}}{\gamma_j^2}\right).$$

In the special case $\alpha = 1/2$, this becomes

$$\|f\|_{K_{1/2,\gamma}}^2 = \|f\|_{L^2}^2 + \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\prod_{j=1}^d \left(1 + \frac{|k_j|}{\gamma_j^2}\right) - 1 \right).$$

Using the identity

$$\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} \frac{|k_j|}{\gamma_j^2} = \prod_{j=1}^d \left(1 + \frac{|k_j|}{\gamma_j^2}\right) - 1,$$

we can express our weighted smoothed-out variation as

$$\sigma_{\text{WSO}}(f)^2 = \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\prod_{j=1}^d \left(1 + \frac{|k_j|}{\gamma_j^2}\right) - 1 \right) = \|f\|_{K_{1/2,\gamma}}^2 - \|f\|_{L^2}^2.$$

Hence, the weighted smoothed-out variation coincides with the Korobov seminorm at smoothness $\alpha = 1/2$. We remark that the recent work [10] provides a universal algorithm for high-dimensional integration for functions in weighted Korobov classes.

3. Constructions for functions with low effective dimension

As discussed in the Introduction, many high-dimensional functions encountered in practice can be well-approximated by depending strongly on only a few coordinates, or on low-order interactions between them. This motivates the notion of effective dimension, which captures the idea that the function's relevant dimensionality is much smaller than its nominal dimension d .

In this section, we introduce two variants of the WSubgTrans algorithm that explicitly incorporate effective dimension. These variants correspond to two standard measures: the *superposition sense*, which quantifies the importance of variable interaction order, and the *truncation sense*, which quantifies the importance of the earliest coordinates. We begin with a brief review of the analysis of variance (ANOVA) decomposition and the associated definitions of effective dimension [24, Appendix A]. Subsection 3.1 addresses the superposition dimension, while Subsection 3.2 focuses on the truncation dimension.

For a subset of coordinates $u \subseteq [d]$, we denote its cardinality by $|u|$, and write $\bar{u} := [d] \setminus u$ for its complement. Given a vector $\mathbf{x} \in \mathbb{R}^d$ and a subset $u \subseteq [d]$, we denote by $\mathbf{x}^u \in \mathbb{R}^{|u|}$ the restriction of $\mathbf{x}$ to the coordinates indexed by u . In the analysis of variance (ANOVA) decomposition, a function f can be expressed as

$$f(\mathbf{x}) = \sum_{u \subseteq [d]} f_u(\mathbf{x}),$$

where each component $f_u(\cdot)$ depends only on the variables $\mathbf{x}^u$. More precisely, for any subset $u \subseteq [d]$, the corresponding term is defined by

$$f_u(\mathbf{x}) := \int_{[0,1]^{\bar{u}}} \left(f(\mathbf{x}) - \sum_{v \subsetneq u} f_v(\mathbf{x}) \right) d\mathbf{x}^{\bar{u}}$$

The zero-order term $f_{\emptyset}(x)$ is a constant that is equal to the mean of f . The ANOVA decomposition is orthogonal in L^2 , so that

$$\sigma(f)^2 = \sum_{u \subseteq [d]} \sigma_u^2, \quad \text{where } \sigma_u^2 := \sigma(f_u)^2.$$

We now recall the definitions of two notions of effective dimension in the superposition sense and in the truncation sense following [5].

Definition 3.1 (Effective dimension, superposition sense). The effective dimension of f , in the superposition sense, is the smallest integer d_s such that $\sum_{0 < |u| \leq d_s} \sigma(f_u)^2 \geq 0.99\sigma(f)^2$.

Definition 3.2 (Effective dimension, truncation sense). The effective dimension of f , in the truncation sense, is the smallest integer d_T that $\sum_{u \subseteq \{1, \dots, d_T\}} \sigma(f_u)^2 \geq 0.99\sigma(f)^2$.

3.1. Effective superposition dimension

For $h \in \mathbb{N}$ and $s \in [d]$, let $\tilde{\mathcal{D}}_{\leq h}^{\otimes s}$ be the collection of dyadic boxes of level at most h that involve at most s non-trivial dimensions; this is a subset of the full dyadic collection $\mathcal{D}_{\leq h}^{\otimes d}$ defined in Subsection 2.1. For functions with small effective dimension in the superposition sense (Definition 3.1), we modify the WSubgTrans algorithm by forming the incidence vector of $\mathbf{z}_j$ with respect to $\tilde{\mathcal{D}}_{\leq h}^{\otimes s}$ rather than the full collection $\mathcal{D}_{\leq h}^{\otimes d}$. Specifically, on line 9 of Algorithm 2, we define the vector $\mathbf{v}_j$ as

$$\mathbf{v}_j = \left(\mathbf{e}_j^{A_T}, (\mathbf{1}_{\{\mathbf{z}_j \in B\}})_{B \in \tilde{\mathcal{D}}_{\leq h}^{\otimes s}} \right), \quad (5)$$

where $\mathbf{1}_{\{\mathbf{z}_j \in B\}} = 1$ if $\mathbf{z}_j \in B$ and 0 otherwise. All other steps of the WSubgTrans algorithm remain unchanged. The modified algorithm outputs n sets of points, each of equal size n ; we use A_T to denote any one of these sets.

Our next result provides a performance guarantee for the point set A_T generated by the modified WSubgTrans algorithm. The proof is given in Appendix A.2.

Theorem 2. For every square-integrable function $f : [0, 1]^d \rightarrow \mathbb{R}$, the integration error satisfies¹

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O_s(1) \cdot \inf_{f=g+h, h \in \mathcal{F}_s} \left(\frac{\log n}{n} \sigma(g)^2 + \frac{d^s (\log(dn))^{s+1}}{n^2} \sigma_{\text{SO}}(h)^2 \right),$$

where the infimum is over all decompositions of f as the sum of two functions $g, h : [0, 1]^d \rightarrow \mathbb{R}$ with $g \in L^2([0, 1]^d)$ and $h \in \mathcal{F}_s$. The class $\mathcal{F}_s$ consists of functions involving interactions of order at most s , namely

$$\mathcal{F}_s = \left\{ \sum_{|u| \leq s} f_u(\mathbf{x}^u) : f_u \in L^2([0, 1]^d) \right\}.$$

The smoothed-out variation is defined as

$$\sigma_{\text{SO}}(h)^2 := \sum_{\mathbf{k} \neq \mathbf{0}} |\hat{h}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} |k_j| \right).$$

Remark 3. In particular, for any function $f : [0, 1]^d \rightarrow \mathbb{R}$ with effective dimension s in the superposition sense, consider its ANOVA decomposition

¹ $O_s(1)$ denotes a constant that depends only on s .

$$f(\mathbf{x}) = \sum_{|u| \leq s} f_u(\mathbf{x}) + \sum_{|u| > s} f_u(\mathbf{x}) =: f_{\leq s} + f_{> s}.$$

Then the integration error satisfies

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O_s \left(\frac{\log n}{n} \sigma(f_{> s})^2 + \frac{d^s (\log(dn))^{s+1}}{n^2} \sigma_{\text{SO}}(f_{\leq s})^2 \right).$$

Since $\sigma(f_{\leq s})^2 \geq 0.99\sigma(f)^2$ and $\sigma(f_{> s})^2 \leq 0.01\sigma(f)^2$, the modified WSubgTrans algorithm is expected to perform well in high dimensions for functions with small effective superposition dimension.

Moreover, the number of nonzero entries in the vector $\mathbf{v}_j$ is $O_s(d^s h^s) = O_s((d \log(dn))^s)$, leading to significantly lower computational cost compared to the full WSubgTrans algorithm described in Section 2. Specifically, the full WSubgTrans algorithm has an overall runtime of

$$O \left(\sum_{t=0}^{T-1} \frac{n^2}{2^t} \cdot h^d \right) = O(n^2 (\log(dn))^d),$$

while the modified algorithm tailored to superposition of order s achieves an overall runtime of

$$O_s \left(\sum_{t=0}^{T-1} \frac{n^2}{2^t} \cdot (d \log(dn))^s \right) = O_s(n^2 (d \log(dn))^s).$$

The corresponding amortized time per output point is $O((\log(dn))^d)$ for the full algorithm and $O_s((d \log(dn))^s)$ for the modified version. When the dimension d is large and the effective superposition dimension s remains moderate, this results in a substantial computational improvement.

3.2. Effective truncation dimension

In this subsection, we modify the WSubgTrans algorithm to accommodate functions with small effective dimension in the truncation sense (Definition 3.2). This setting can be viewed as a special case of the general WSubgTrans algorithm presented in Section 2, where the weights are chosen as $\gamma_1 = \dots = \gamma_s = 1$ and $\gamma_{s+1} = \dots = \gamma_d = 0$ for a fixed integer $s \in [d]$. In this case,

$$\|\mathbf{v}_j\|_2^2 = 1 + \prod_{j \in [d]} (1 + h\gamma_j^2) = O(h^s) = O((\log(dn))^s).$$

The overall runtime of the algorithm with this specific choice of weights is $O(n^2 (\log(dn))^s)$, yielding an amortized time per output point of $O((\log(dn))^s)$.

Our next result provides a performance guarantee for the point set A_T generated by the modified WSubgTrans algorithm. The proof is given in Appendix A.2.

Theorem 3. For every square-integrable function $f : [0, 1]^d \rightarrow \mathbb{R}$, the integration error satisfies

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O_s(1) \cdot \inf_{f=g+h, h \in \tilde{\mathcal{F}}_s} \left(\frac{\log n}{n} \sigma(g)^2 + \frac{(\log(dn))^{s+1}}{n^2} \sigma_{\text{SO}}(h)^2 \right),$$

where the infimum is over all decompositions of f as the sum of two functions $g, h : [0, 1]^d \rightarrow \mathbb{R}$ with $g \in L^2([0, 1]^d)$ arbitrary and $h \in \tilde{\mathcal{F}}_s$. The class $\tilde{\mathcal{F}}_s$ consists of functions that depend only on the variables $\mathbf{x}^{(1,2,\dots,s)}$.

Remark 4. For any function $f : [0, 1]^d \rightarrow \mathbb{R}$ with effective dimension s in the truncation sense, consider its ANOVA decomposition

$$f(\mathbf{x}) = \sum_{u \subseteq \{1,2,\dots,s\}} f_u(\mathbf{x}) + \sum_{u \not\subseteq \{1,2,\dots,s\}} f_u(\mathbf{x}) =: f_s^{(1)} + f_s^{(2)}.$$

Table 1

Star discrepancy: Sobol' and IID baseline and SubgTrans mean (ST) with different initializations of distribution and population size.

n	Sobol'	IID	ST (IID)		ST (Sobol')	
			n^2	$16n$	n^2	$16n$
8	0.312500	0.354866	0.301768	0.325006	0.305120	0.318114
16	0.171875	0.266701	0.199798	0.190557	0.201280	0.206695
32	0.089844	0.182864	0.140638	0.149751	0.125465	0.120430
64	0.053711	0.148777	0.084015	0.093255	0.088080	0.084042
128	0.025146	0.106092	0.053085	0.059950	0.053417	0.051763
256	0.014587	0.082900	0.033081	0.034835	0.032951	0.032138

Then the integration error satisfies

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O_s \left(\frac{\log n}{n} \sigma(f_s^{(2)})^2 + \frac{(\log(dn))^{s+1}}{n^2} \sigma_{\text{SO}}(f_s^{(1)})^2 \right).$$

Since $\sigma(f_s^{(1)})^2 \geq 0.99\sigma(f)^2$ and $\sigma(f_s^{(2)})^2 \leq 0.01\sigma(f)^2$, the modified WSubgTrans algorithm is expected to perform well in high dimensions for functions with small effective truncation dimension.

4. Numerical experiments

We evaluate four variants of the Subgaussian Transference algorithm: the original formulation from [3], Algorithm 2 from Section 2, and the two variants from Section 3 designed to account for effective dimension in the truncation and superposition senses. Our numerical study examines both the uniformity of the resulting point sets and their performance in integral approximation.

4.1. Implementation detail

To reduce the algorithm runtime, several practical modifications were introduced in our implementations. First, the parameter λ was set sufficiently small (e.g., $\lambda \approx 10^{-3}$) in Algorithm 1, and instead of failing, the algorithm chooses the sign c_j greedily when $|\langle \mathbf{w}_{j-1}, \mathbf{v}_j \rangle| > \lambda$. Second, instead of drawing n^2 candidate points, we use a reduced population of size kn , with k a power of two (usually $k = 16$). This lowers both time and memory complexity to $O(n)$. When $k = n$, we recover the precise oversampling procedure in the algorithm. When an effective dimension s is specified, the dyadic refinement parameter is set to $h = \lceil \log_2(sn) \rceil$ rather than the nominal choice $h = \lceil \log_2(dn) \rceil$, capping the exponential growth of dyadic boxes in high dimensions. Lastly, we experiment with initializing the algorithm with the first kn Sobol' points, rather than uniform IID random points to investigate the effect of the values of the starting set.

These cost-saving adjustments do come at the expense of some algorithmic precision—most notably due to the coarser dyadic refinement and smaller oversampling initialization. However, as we will see, they do not compromise performance enough to prevent improvements over MC methods.

4.2. Resulting star discrepancy

We implement the original SubgTrans algorithm from [3] (or equivalently, WSubgTrans with unit weights) for each $n \in \{8, 16, 32, 64, 128, 256\}$ and $d = 2$. We generate output sets from population sizes of kn (i) $k = 16$, and (ii) $k = n$ to test the algorithm for robustness under the size of initialization set. To assess the geometric uniformity of the resultant sets and to illustrate the transference principle (see Lemma 4), we compute the star discrepancy (2) for each output set and report the mean for each configuration. As reference, we also report the star discrepancy of the unscrambled Sobol' set and the average of 16 independent IID uniform point sets of the same n .

From the results contained in Table 1, Sobol' generally attains the lowest star discrepancy for $n \geq 16$. For example, at $n = 128$, the Sobol' set reports a star discrepancy of 0.025, whereas SubgTrans

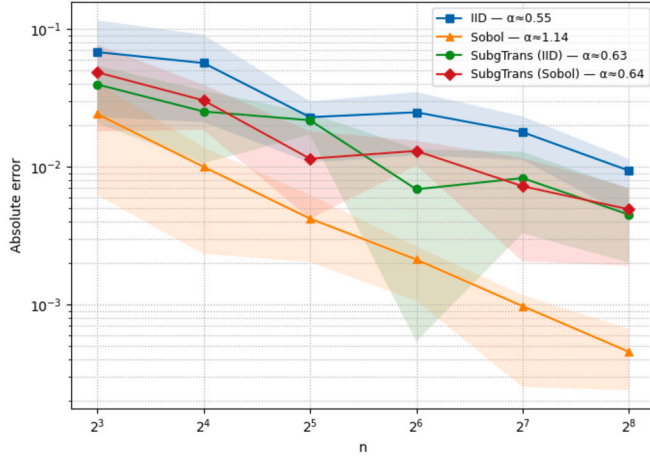


Fig. 2. MAE with IQR for the approximation of the integral of function f using WSubgTrans (with IID and Sobol' initializations), Sobol' and IID.

averages range from 0.052 to 0.060 depending on initialization. However, for $n = 128$, we found in our investigation that the minimum star discrepancy reported was 0.040697 for Sobol' initialization and $k = n$, and 0.039367 for IID initialization and $k = n$. This brings us to our key point: initialization matters; with Sobol' starts, $k = 16$ tends to slightly beat the $k = n$ start for $n \geq 32$; with uniform starts, the $k = n$ start tends to be better at larger n .

We also note that SubgTrans with IID initialization is always superior in comparison to IID. This suggests that in applications where only IID samples are available (and thus Sobol' is not applicable), subgaussian transference provides a competitive alternative to MC.

4.3. Small truncation dimension

We now move to the high-dimensional integration setting, restricting attention to problems of low effective dimension due to computational constraints. To test the truncation variant of WSubgTrans introduced in Section 3.2, we consider the function

$$f(\mathbf{x}) = \sum_{i=1}^d (-1)^i \prod_{j=1}^i x_j$$

which for nominal dimension $d = 100$ is known to have a truncation effective dimension of 2 [17]. The true integral value can be calculated analytically as $\frac{(1 - (-1/2)^d)}{-3}$. We approximate this integral using sampling nodes generated by WSubgTrans with weight vector $(1, 1, 0, \dots, 0)$, alongside IID and scrambled Sobol' baselines, and compute the absolute error to the known true value. The convergence rates are also reported in the form $\mathcal{O}(n^{-\alpha})$.

Fig. 2 shows that the Sobol' sequence provides the strongest performance, with rapid decay of error and low variability across replications. Nevertheless, WSubgTrans delivers a clear improvement over IID sampling, achieving faster convergence ($\alpha \approx 0.65$ vs. $\alpha \approx 0.55$ for IID) and consistently smaller absolute errors. It seems initialization has little effect in this example: whether starting from IID or Sobol' inputs, the partitioned sets exhibit nearly identical accuracy, indicating that the algorithm robustly adapts to the effective dimension structure. This highlights the central benefit of WSubgTrans: even in very high nominal dimension ($d = 100$), exploiting the effective dimension knowledge of a problem and restricting the action to the important variables yields point sets that outperform IID by imposing low-discrepancy structure where it matters.

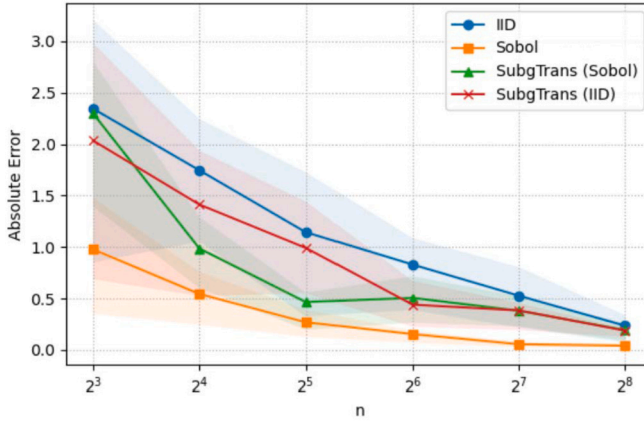


Fig. 3. Approximation error of an Asian call option pricing problem using WSubgTrans (with IID and Sobol' initializations), Sobol', and IID.

4.4. Small superposition dimension

As a benchmark application with low effective dimension, we consider pricing an Asian call option on a stock following geometric Brownian motion; see [33,19,9]. The option payoff depends on the average of the asset price observed at monthly intervals over one year, leading to a $d = 12$ dimensional integration problem. The present value is given by

$$C_0 = \mathbb{E} \left[e^{-rT} \left(\frac{1}{d} \sum_{j=1}^d S(u_j) - K \right)^+ \right]. \quad (6)$$

where $S(u)$ is the stock price at time u under the risk-neutral measure, K is the strike, T the maturity, and r the risk-free rate. Assuming geometric Brownian motion with volatility σ , this expectation admits an integral representation over $[0, 1]^d$ in terms of inverse normal transforms of uniform random variables [9]. Using parameters $S(0) = 50$, $K = 45$, $T = 1$, $r = 0.05$, and $\sigma = 0.3$, the true option value is $C_0 = 7.2110915$, estimated in advance to high precision using 2M Sobol' points.

Like many finance problems, this integral has a much lower effective dimension than its nominal value: the superposition effective dimension is approximately 2, so most of the variance is explained by one- and two-way interactions among inputs. To exploit this, we apply the WSubgTrans algorithm with unit coordinate weights with $s = 2$, i.e., the algorithm will only construct the incidence vector with $\tilde{\mathcal{D}}_{\leq h}^{\otimes 2}$, dyadic intervals up to and including order 2. We generate sets of size $n \in \{8, 16, 32, 64, 128, 256\}$ from initial populations of size $16n$. We again compare against IID and scrambled Sobol' points.

We see from Fig. 3 that as per the previous example, Sobol' still provides the sharpest convergence, but WSubgTrans tracks much more closely to Sobol' for small and moderate n , narrowing the gap substantially relative to IID. The shaded error bands reveal that WSubgTrans points also reduce variability across repetitions, producing more stable estimates than IID even at coarse sample sizes. Once again, we see that by partitioning the population set via subgaussian transference in a targeted manner to obtain sampling nodes, when something is known about the problem, one can achieve much better results than random sampling.

5. Discussion

In this work we extended the original subgaussian transference algorithm of [3] to the weighted setting, as is standard in modern QMC methods, and introduced effective-dimension variants targeting

both truncation and superposition structure. On the theoretical side, we extended the error analysis in [3] to capture the weighted and low effective-dimension structure. This provides error guarantees that depend explicitly on the chosen weights, and hence on the effective dimension of the integrand, rather than on the full Hardy–Krause variation as in the classical Koksma–Hlawka inequality.

The algorithms developed here are also related to the *subset selection* problem [6], which seeks a single subset minimizing star discrepancy. In contrast, subgaussian transference partitions a large pool into many n -point sets, naturally providing replications that are useful in error approximation.

On the computational side, our modifications of reducing candidate pool size, adjusting refinement depth, and setting λ small, kept run-times tractable although paying a cost of accuracy at times. Generally speaking, scaling remains an issue: the exponential growth of dyadic boxes in the full algorithm makes naive application infeasible in even moderate dimensions. The effective-dimension variants alleviate this to some degree by reducing the combinatorial complexity, but further work is needed to bring the approach to genuinely high dimensional problems.

Overall, subgaussian transference provides theoretically grounded guarantees for integration in weighted spaces, alongside an algorithmic framework that produces replicated QMC point sets adapted to effective dimension. In all of the numerical experiments, subgaussian transference (with IID initializations) outperforms MC, providing a more favorable alternative in situations where only random samples are accessible. We have also seen that it does not yet outperform the standard Sobol' QMC construction in practice and further work is required to make the algorithm competitive.

Acknowledgements

The third author was supported by NSF grant DMS-2316011 while affiliated with Illinois Institute of Technology. The authors would like to thank Zexin Pan, Art B. Owen, Christiane Lemieux and Fred J. Hickernell for helpful discussions. Python code for SubgTrans and WSubgTrans can be found at [14].

Appendix A

A.1. Proofs of results in Section 2

Notation. For integer $d > 0$, we write $[d] := \{1, \dots, d\}$. We use bold lowercase letters, e.g., $\mathbf{z} = (z_1, \dots, z_d)$, to denote vectors, with z_j denoting the j^{th} coordinate of $\mathbf{z}$. For a collection of vectors, we denote the j^{th} vector by $\mathbf{z}_j$, and refer to its k^{th} coordinate as $z_{j,k}$. For a subset of coordinates $u \subseteq [d]$, we denote its cardinality by $|u|$, and write $\bar{u} := [d] \setminus u$ for its complement. Given a vector $\mathbf{x} \in \mathbb{R}^d$ and a subset $u \subseteq [d]$, we denote by $\mathbf{x}^u \in \mathbb{R}^{|u|}$ the restriction of $\mathbf{x}$ to the coordinates indexed by u .

Dyadic Intervals. Let $j \in \mathbb{N}$, we denote by $\mathcal{D}_j$ the set of all left-open dyadic intervals in $(0, 1]$ of length 2^{-j} , i.e., $\mathcal{D}_j := \{(\ell/2^j, (\ell+1)/2^j] : \ell = 0, \dots, 2^j - 1\}$, and call j the level of these dyadic intervals. For $h \in \mathbb{N}$, define $\mathcal{D}_{\leq h} := \bigcup_{j \leq h} \mathcal{D}_j$ as the set of all dyadic intervals of level less than or equal to h . In higher dimensions, $\mathcal{D}_{\leq h}^{\otimes d} := \{I_1 \times \dots \times I_d : I_j \in \mathcal{D}_{\leq h} \text{ for all } j \in [d]\}$ denotes all d -dimensional dyadic boxes with level at most h in each dimension.

Prefix Intervals. For any $x \in [0, 1]$, we denote left-open prefix interval $C_x := (0, x]$, with $C_1 = (0, 1]$. For any $\mathbf{z} = (z_1, \dots, z_d) \in [0, 1]^d$, we denote $C_{\mathbf{z}} := (0, z_1] \times \dots \times (0, z_d]$ the left-open corner at $\mathbf{z}$. We also denote $C^d := \{C_{\mathbf{z}} : \mathbf{z} \in [0, 1]^d\}$ the set of all left-open corners. For $h \in \mathbb{N}$, we use C_h to denote all left-open prefix intervals whose length is an integer multiple of 2^{-h} (excluding² $(0, 1]$), i.e. $C_h := \{C_{\ell/2^h} : \ell = 0, \dots, 2^h - 1\}$. We use $C_h^{\otimes d} := \{I_1 \times \dots \times I_d : I_j \in C_h \text{ for all } j \in [d]\}$ to denote all left-open corners at grid points in $2^{-h}\mathbb{Z}^d \cap [0, 1)^d$.

Dyadic Decomposition. *Dyadic decomposition* refers to decomposing sub-intervals of $(0, 1]$ into a minimal disjoint union of dyadic intervals. We only use dyadic decomposition for prefix intervals. For any $h \in \mathbb{N}$, we define $P_h \in \{0, 1\}^{|C_h| \times (|\mathcal{D}_{\leq h}| - 1)}$ the dyadic decomposition matrix of C_h , where the

² This is again due to the slight technicality caused by the boundary.

number of columns is $|\mathcal{D}_{\leq h}| - 1$ because we have chosen to exclude the unused $(0, 1] \in \mathcal{D}_{\leq h}$ in the decomposition. In particular, for the ℓ th prefix interval $I_\ell \in \mathcal{C}_h$ and the r th dyadic interval $J_r \in \mathcal{D}_{\leq h} \setminus \{C_1\}$, we have $(P_h)_{\ell,r} = 1$ if J_r is used in the dyadic decomposition of I_ℓ , and 0 otherwise.

As each prefix in $\mathcal{C}_{\leq h}$ is a sum of at most h dyadic intervals in $\mathcal{D}_{\leq h}$, each row of P_h has at most h ones. However, the columns of P_h could be dense with $\Omega(2^h)$ entries. For our analysis, we will in fact crucially exploit the structure of the columns of P_h . For technical convenience and following [3, Lemma 2.2], we actually work with a slight variant of P_h that has more structured columns, which we call the *structured decomposition matrix* and denote it as $\bar{P}_h \in \{0, 1\}^{|\mathcal{C}_h| \times (|\mathcal{D}_{\leq h}| - 1)}$. In higher dimension, the dyadic decomposition of $\mathcal{C}_h^{\otimes d}$ into dyadic boxes $\mathcal{D}_{\leq h}^{\otimes d}$ is given by $P_h^{\otimes d}$, and the structured decomposition tensor is given by $(\bar{P}_h)^{\otimes d}$.

A.1.1. Combinatorial discrepancy, transference principle, and subgaussianity

Let us consider an iteration $t \in \{0, \dots, T-1\}$ of the WSubgTrans algorithm where A_{t+1} is produced from A_t by running BalSubgDisc. For any set $C \subseteq [0, 1]^d$, let

$$\text{disc}_t(C) := |\{\mathbf{z}_j \in C \cap A_t \text{ with } c_{t,j} = +1\}| - |\{\mathbf{z}_j \in C \cap A_t \text{ with } c_{t,j} = -1\}| \quad (\text{A.1})$$

denote the *combinatorial discrepancy* of $C \cap A_t$. In particular, we denote $\text{disc}_t(\mathbf{z}) := \text{disc}_t(C_{\mathbf{z}})$ for any $\mathbf{z} \in [0, 1]^d$. For each $t \in \{0, \dots, T\}$, we denote $n_t := |A_t| = 2^{-t}n_0$. For any measurable set $C \subseteq [0, 1]^d$, denote $n_t(C) := |C \cap A_t|$, and *continuous discrepancy* $h_t(C) := \text{vol}(C) - n_t(C)/n_t$. We denote $h_t(\mathbf{z}) := h_t(C_{\mathbf{z}})$ for any $\mathbf{z} \in [0, 1]^d$.

Let $\mathcal{I} := \mathcal{D}_{\leq h}^{\otimes d} \setminus \{C_1\}$ be the collection of left-open dyadic boxes up to level h in dimension d , excluding the box C_1 . Define the (unweighted) combinatorial discrepancy vector $\mathbf{d}^{t,\mathcal{D}} \in \mathbb{Z}^{|\mathcal{I}|}$ by

$$(\mathbf{d}^{t,\mathcal{D}})_B := \text{disc}_t(B), \quad B \in \mathcal{I}.$$

Define the weighted combinatorial discrepancy vector $\mathbf{wd}^{t,\mathcal{D}} \in \mathbb{R}^{|\mathcal{I}|}$ by

$$(\mathbf{wd}^{t,\mathcal{D}})_B := \text{wdisc}_t(B) := \gamma(B) \text{disc}_t(B), \quad B \in \mathcal{I},$$

where $\gamma(B) = \prod_{j \in [d]: B_j \neq (0,1]} \gamma_j$ is the product weight defined in (4).

The continuous discrepancy and combinatorial discrepancy are tightly connected through the transference principle, which enables the construction of low continuous discrepancy sets by recursively applying colorings that minimize combinatorial discrepancy. This connection is made precise in [3, Lemma 3.4]:

Lemma 4 (Transference principle, [3, Lemma 3.4]). *For any measurable set $C \subseteq [0, 1]^d$, we have*

$$h_T(C) = h_0(C) + \sum_{t=0}^{T-1} \frac{\text{disc}_t(C)}{n_t}.$$

Remark 5 (Transference principle). The identity in Lemma 4 shows that the continuous discrepancy at the final level is obtained by accumulating, across all levels, the normalized combinatorial discrepancies produced by the successive colorings. At level t , the coloring splits the points in A_t into two classes, and A_{t+1} is obtained by retaining one of them. Thus, whenever the coloring is imbalanced inside a set C , the empirical mass of C after halving deviates from the ideal factor $1/2$, and this deviation is exactly measured by $\text{disc}_t(C)/n_t$. For completeness, we give a short proof of Lemma 4. Since

$$h_t(C) = \text{vol}(C) - \frac{n_t(C)}{n_t},$$

we may write

$$h_T(C) - h_0(C) = \sum_{t=0}^{T-1} (h_{t+1}(C) - h_t(C)) = \sum_{t=0}^{T-1} \left(\frac{n_t(C)}{n_t} - \frac{n_{t+1}(C)}{n_{t+1}} \right).$$

Using $n_t = 2n_{t+1}$, we obtain

$$\frac{n_t(C)}{n_t} - \frac{n_{t+1}(C)}{n_{t+1}} = \frac{|C \cap A_t| - 2|C \cap A_{t+1}|}{n_t}.$$

Since A_{t+1} is formed by keeping one color class from the coloring of A_t , the quantity

$$|C \cap A_t| - 2|C \cap A_{t+1}|$$

is exactly the signed imbalance of the coloring on $C \cap A_t$, namely $\text{disc}_t(C)$. Therefore,

$$h_T(C) = h_0(C) + \sum_{t=0}^{T-1} \frac{\text{disc}_t(C)}{n_t}.$$

Example 1 (One-dimensional illustration). Consider $d = 1$ and $C = [0, z] \subseteq [0, 1]$. Suppose

$$A_t = \{x_1, x_2, x_3, x_4\}, \quad n_t = 4,$$

with

$$x_1, x_2, x_3 \in C, \quad x_4 \notin C.$$

Assume that the coloring on A_t assigns signs

$$c_{t,1} = +1, \quad c_{t,2} = -1, \quad c_{t,3} = -1,$$

to the three points in C , and that A_{t+1} is formed by retaining the points colored -1 . Then

$$\text{disc}_t(C) = 1 - 2 = -1, \quad n_t(C) = 3, \quad n_{t+1}(C) = 2.$$

Since $n_{t+1} = 2$, we compute

$$\frac{n_t(C)}{n_t} - \frac{n_{t+1}(C)}{n_{t+1}} = \frac{3}{4} - \frac{2}{2} = -\frac{1}{4} = \frac{\text{disc}_t(C)}{n_t}.$$

This shows concretely how the coloring imbalance at level t produces the increment in continuous discrepancy between levels t and $t + 1$.

By Lemma 4, we can control the integration error using the final point set A_T by bounding the error at each intermediate step $t \in \{0, \dots, T - 1\}$.

Moreover, the WSubgTrans algorithm ensures subgaussianity of both the coloring vector $\mathbf{c}_t$ and the weighted combinatorial discrepancy vector $\mathbf{wd}^{t,\mathcal{D}}$ as follows:

Lemma 5 (Subgaussianity for dyadic boxes). For any $0 \leq t \leq T - 1$, WSubgTrans satisfies (1) $\text{disc}_t([0, 1]^d) = 0$, and (2) the random vector $(\mathbf{c}_t, \mathbf{wd}^{t,\mathcal{D}})$ is $O(d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n))$ -subgaussian.³

Proof. The first claim follows because BalSubgDisc always produces a balanced coloring.

The subgaussian property is based on the following result for the BalSubgDisc algorithm:

³ A random vector $\mathbf{y} \in \mathbb{R}^m$ is called σ^2 -subgaussian if for all $\mathbf{z} \in \mathbb{R}^m$, one has $\mathbb{E}[\exp(\langle \mathbf{y}, \mathbf{z} \rangle)] \leq \exp(\sigma^2 \|\mathbf{z}\|_2^2 / 2)$.

Theorem 6 ([2]). Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^m$ with $\|\mathbf{v}_j\|_2 \leq 1$ online, the Self-Balancing Walk computes a symmetric random coloring $\mathbf{c} = (c_1, \dots, c_n) \in \{\pm 1\}^n$, such that $\sum_{j=1}^t c_j \mathbf{v}_j$ is $O(\log(mn))$ -subgaussian for all $t \in [n]$ whp. Moreover, the algorithm runs in $O(\sum_{j \in [n]} \text{nnz}(\mathbf{v}_j))$ time.⁴

For every point $\mathbf{z}_j \in A_t$ and every subset $u \subseteq [d]$, the point $\mathbf{z}_j$ lies in exactly $h^{|u|}$ dyadic boxes in $\mathcal{D}_{\leq h}^{\otimes d}$ whose set of the nontrivial coordinate directions is precisely u . Thus

$$\begin{aligned} \|\mathbf{v}_j\|_2^2 &= 1 + \sum_{B \in \mathcal{D}_{\leq h}^{\otimes d}} w^2(B) \mathbf{1}_{\{\mathbf{z}_j \in B\}} \\ &= 1 + \sum_{B \in \mathcal{D}_{\leq h}^{\otimes d}} \mathbf{1}_{\{\mathbf{z}_j \in B\}} \prod_{j \in [d]: B_j \neq (0,1)} \gamma_j^2 \\ &= 1 + \sum_{u \subseteq [d]} h^{|u|} \prod_{j \in u} \gamma_j^2 \\ &= 1 + \prod_{j \in [d]} (1 + h\gamma_j^2), \end{aligned}$$

and the claim follows from Theorem 6 on the at most n^2 vectors $\mathbf{v}_j$, $m = O(2^{(h+1)d})$ and $h = O(\log(dn))$. $\square$

A.1.2. Proof of Theorem 1

As noted in [3, Remark 1.1], requiring that f has a Fourier series uses some (mild) smoothness assumptions, and it also requires 1-periodicity, but this is without loss of generality via standard folding. These can be removed by exploiting that our method uses random samples and that the Fourier series of f converges to f in a ℓ_2 sense. We ignore these technicalities as they are standard and do not give any new insight.

Overview of proof of Theorem 1 Before turning to the details of the proof of Theorem 1, we briefly summarize its main steps. We begin with the Hlawka–Zaremba formula, which expresses the integration error in terms of the continuous discrepancy of the output point set. We then apply the transference principle (Lemma 4) to decompose the final continuous discrepancy into the initial sampling error and the discrepancy increments generated at each halving step of the algorithm. This reduces the proof to controlling the contribution from each level t .

The initial term is simply the Monte Carlo error of the n^2 IID samples. For the discrepancy terms, we combine the subgaussian control provided by the balanced coloring procedure with the dyadic decomposition to reduce the problem to bounding a suitable weighted ℓ_2 quantity; see (A.7). We then estimate this quantity via a Fourier-analytic argument.

There are several additional technical points. First, while the Fourier basis is orthogonal, this orthogonality is no longer directly available after applying the operator $\overline{P}_h^\top$ (and its tensorized version). To overcome this, we introduce a random shift of the dyadic system. This shift modifies the phases of the Fourier coefficients without changing their magnitudes, and after averaging over the shift, orthogonality is restored so that the cross terms vanish; see Corollary 1. Second, in higher dimensions, we exploit the natural tensor-product structure of the one-dimensional dyadic decomposition. The higher-dimensional Hlawka–Zaremba formula also produces additional boundary terms, which are handled by exploiting the joint subgaussianity on the corresponding boundaries and carrying out a similar argument there.

We are now ready to prove Theorem 1.

⁴ $\text{nnz}(\mathbf{z})$ denotes number of non-zero coordinates in a vector $\mathbf{z}$.

Proof of Theorem 1. Fix an arbitrary 1-periodic function $f \in L^2([0, 1]^d)$ with a continuous mixed derivative. We can equivalently view WSubgTrans as fixing $\mathcal{D}_{\leq h}^{\otimes d}$ and applying a random shift $\mathbf{s} \sim [0, 1]^d$ to f which results in $f_{\mathbf{s}}(\mathbf{z}) = f(\mathbf{z} + \mathbf{s})$. Denote $\mathbb{E}_{\mathbf{s}}$ the expectation w.r.t. random shift $\mathbf{s}$. By Hlawka-Zaremba formula, the integration error using point set A_T is

$$\text{err}(A_T, f_{\mathbf{s}}) = \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u f_{\mathbf{s}}(\mathbf{z}) d\mathbf{z},$$

where $h_T(\mathbf{z}) = \text{vol}(C_{\mathbf{z}}) - |C_{\mathbf{z}} \cap A_T|/n_T$ is the (scaled) continuous discrepancy of the corner $C_{\mathbf{z}}$, $Q_u := \{\mathbf{z} \in [0, 1]^d : z_j = 1 \text{ for all } j \notin u\}$ denotes the (shifted) $|u|$ -dimensional cube in $\mathbb{R}^d$, and $\partial_u f_{\mathbf{s}}(\mathbf{z}) = \frac{\partial^{|u|} f_{\mathbf{s}}(\mathbf{z})}{\partial \mathbf{z}_u}$. By Lemma 4, we can write

$$h_T(\mathbf{z}) = h_0(\mathbf{z}) + \sum_{t=0}^{T-1} \frac{\text{disc}_t(\mathbf{z})}{n_t}.$$

Substituting this expression into the error representation yields

$$\begin{aligned} \text{err}(A_T, f_{\mathbf{s}}) &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u f_{\mathbf{s}}(\mathbf{z}) d\mathbf{z} \\ &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} h_0(\mathbf{z}) \cdot \partial_u f_{\mathbf{s}}(\mathbf{z}) d\mathbf{z} \end{aligned} \quad (\text{A.2})$$

$$\begin{aligned} &+ \sum_{t=0}^{T-1} \left(\sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} \frac{\text{disc}_t(\mathbf{z})}{n_t} \cdot \partial_u f_{\mathbf{s}}(\mathbf{z}) d\mathbf{z} \right) \\ &=: \text{err}(A_0, f_{\mathbf{s}}) + \sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_{\mathbf{s}}). \end{aligned} \quad (\text{A.3})$$

First, since A_0 consists of n^2 independent random samples,

$$\mathbb{E}[(\text{err}(A_0, f_{\mathbf{s}}))^2] \leq \frac{\sigma(f)^2}{n^2}. \quad (\text{A.4})$$

We next analyze the discrepancy error $\text{err}_t^{\text{disc}}$ in (A.2). Observe that

$$\begin{aligned} \text{err}_t^{\text{disc}}(f_{\mathbf{s}}) &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} \frac{\text{disc}_t(\mathbf{z})}{n_t} \cdot \partial_u f_{\mathbf{s}}(\mathbf{z}) d\mathbf{z} \\ &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} \frac{\gamma(u) \text{disc}_t(\mathbf{z})}{n_t} \cdot \frac{\partial_u f_{\mathbf{s}}(\mathbf{z})}{\gamma(u)} d\mathbf{z} \\ &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} \frac{\text{wdisc}_t(C_{\mathbf{z}})}{n_t} \cdot \frac{\partial_u f_{\mathbf{s}}(\mathbf{z})}{\gamma(u)} d\mathbf{z}, \end{aligned} \quad (\text{A.5})$$

where $\gamma(u) = \prod_{j \in u} \gamma_j$ is the product weight.

At a high level, we analyze the contribution from each $\emptyset \neq u \subseteq [d]$ using tensorization, and then exploit the joint subgaussianity of discrepancy on each Q_u to combine them.

Let us fix $\emptyset \neq u \subseteq [d]$ and consider the integral in (A.5) for u . Let us assume wlog that the combinatorial discrepancy $\text{disc}_t(\mathbf{z})$ on Q_u is a step function with values changing only when any

coordinate reaches $2^{-h}\mathbb{Z} \cap [0, 1)$. This way, we may view $\text{disc}_t(\mathbf{z})$ restricted to Q_u as a vector⁵ $\mathbf{d}^{t,C,u} \in \mathbb{Z}^{|C_h^{\otimes u}|}$ with coordinates⁶ $\mathbf{d}_j^{t,C,u} = \text{disc}_t(\mathbf{j}/2^h)$ for all $\mathbf{j} \in \{0, \dots, 2^h - 1\}^u$ corresponding to corners $C_h^{\otimes u}$. Similarly, we may view $\text{wdisc}_t(\mathbf{z})$ restricted to Q_u as a vector $\mathbf{wd}^{t,C,u} \in \mathbb{Z}^{|C_h^{\otimes u}|}$ with coordinates $\mathbf{wd}_j^{t,C,u} = \text{wdisc}_t(\mathbf{j}/2^h)$ for all $\mathbf{j} \in \{0, \dots, 2^h - 1\}^u$ corresponding to corners $C_h^{\otimes u}$.

We also define the vector $\mathbf{y}^{f_s,u} \in \mathbb{R}^{|C_h^{\otimes u}|}$ as

$$\mathbf{y}_j^{f_s,u} := \int_{[\mathbf{j}/2^h, (\mathbf{j}+\mathbf{1}_u)/2^h]} \partial_u f_s(\mathbf{z}^u, \mathbf{1}_{\bar{u}}) \cdot d\mathbf{z}^u,$$

for all $\mathbf{j} \in \{0, \dots, 2^h - 1\}^u$. It is possible to further express $\mathbf{y}_j^{f_s,u}$ as a sum of function values at the vertices of $[\mathbf{j}/2^h, (\mathbf{j}+\mathbf{1}_u)/2^h] \times \mathbf{1}_{\bar{u}}$ with mixed signs, but we do not need such a formula. Next, observe that the vector $\mathbf{y}^{f_s,u}$ takes a tensor form if the function f_s is the product of functions of each coordinate.

Observation 1 (*Tensor form for product functions*). If $f_s : [0, 1]^d \rightarrow \mathbb{C}$ has the form $f_s(\mathbf{z}) = \prod_{i \in u} g_i(z_i)$ for all $\mathbf{z} \in Q_u$, then we have $\mathbf{y}_j^{f_s,u} = \prod_{i \in u} \mathbf{y}_{j_i}^{g_i}$ for any $\mathbf{j} \in \{0, \dots, 2^h - 1\}^u$, where $\mathbf{y}^{g_i}$ is defined as in [3]. Consequently, we have $\mathbf{y}^{f_s,u} = \otimes_{i \in u} \mathbf{y}^{g_i}$.

This observation is immediate from the definition of $\mathbf{y}^{f_s,u}$ above. Using dyadic decomposition, we may now express $\text{err}_t^{\text{disc}}$ as inner products

$$\begin{aligned} \text{err}_t^{\text{disc}}(f_s) &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} \frac{\text{wdisc}_t(C_{\mathbf{z}})}{n_t} \cdot \frac{\partial_u f_s(\mathbf{z})}{\gamma(u)} d\mathbf{z} \\ &= \frac{1}{n_t} \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \cdot \left\langle \mathbf{wd}^{t,C,u}, \frac{\mathbf{y}^{f_s,u}}{\gamma(u)} \right\rangle \\ &= \frac{1}{n_t} \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \cdot \left\langle \mathbf{wd}^{t,\mathcal{D},u}, (P_h^\top)^{\otimes u} \frac{\mathbf{y}^{f_s,u}}{\gamma(u)} \right\rangle, \end{aligned} \quad (\text{A.6})$$

where $\mathbf{wd}^{t,\mathcal{D},u} \in \mathbb{Z}^{(|\mathcal{D}_{\leq h} \setminus C_1|)^{\otimes u}}$ is the restriction of $\mathbf{wd}^{t,\mathcal{D}}$ to dyadic boxes in Q_u excluding those with any dimension in u being $(0, 1]$. It follows that the vectors $\mathbf{wd}^{t,\mathcal{D},u}$ for different $\emptyset \neq u \subseteq [d]$ correspond to disjoint coordinates of $\mathbf{wd}^{t,\mathcal{D}}$. Consequently, by Lemma 5, for any outcome of the random shift $\mathbf{s}$, $\text{err}_t^{\text{disc}}(f_s)$ is also subgaussian with parameter

$$\begin{aligned} &O\left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n_t^2} \cdot \sum_{\emptyset \neq u \subseteq [d]} \frac{1}{\gamma(u)^2} \|(P_h^\top)^{\otimes u} \mathbf{y}^{f_s,u}\|_2^2\right) \\ &\leq O\left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n_t^2} \cdot \sum_{\emptyset \neq u \subseteq [d]} \frac{1}{\gamma(u)^2} \|(\bar{P}_h^\top)^{\otimes u} \mathbf{y}^{f_s,u}\|_2^2\right). \end{aligned} \quad (\text{A.7})$$

Therefore, we are left to derive an upper bound on $\sum_{\emptyset \neq u \subseteq [d]} \frac{1}{\gamma(u)^2} \|(\bar{P}_h^\top)^{\otimes u} \mathbf{y}^{f_s,u}\|_2^2$, for which we look at the Fourier transformation of f_s .

⁵ We have ignored the 0-measure set of corners $C \subseteq Q_u$ with some dimension being $(0, 1]$.

⁶ Strictly speaking, the index $\mathbf{j}$ here should really range over $\{0, \dots, 2^h - 1\}^u \times \{2^h\}^{\bar{u}}$ so that the coordinates in $\bar{u}$ of $\mathbf{j}/2^h$ are 1. But to keep notation clear, even though we are working with d -dimensional vectors we only display the coordinates of $\mathbf{j}$ in u which are the only ones that can vary. The same convention applies when we talk about corners $C_h^{\otimes u}$, dyadic boxes $\mathcal{D}_{\leq h}^{\otimes u}$, and decomposition $P_h^{\otimes u}$.

Bounding the ℓ_2 Norm via Fourier. We may write

$$f_s(\mathbf{z}) = \sum_{\mathbf{k} \in \mathbb{Z}^d} \widehat{f}_s(\mathbf{k}) \cdot e^{2\pi i \langle \mathbf{k}, \mathbf{z} \rangle} =: \sum_{\mathbf{k} \in \mathbb{Z}^d} \widehat{f}_s(\mathbf{k}) \cdot e_{\mathbf{k}}(\mathbf{z}),$$

where $e_{\mathbf{k}}(\mathbf{z}) := \exp(2\pi i \langle \mathbf{k}, \mathbf{z} \rangle)$. Again, by replacing f_s with $f_s - \widehat{f}(\mathbf{0})$, we may assume wlog that $\widehat{f}_s(\mathbf{0}) = 0$. Then we may write $\mathbf{y}^{f_s, u} = \sum_{\mathbf{k} \in \mathbb{Z}^d \setminus \mathbf{0}} \widehat{f}_s(\mathbf{k}) \cdot \mathbf{y}^{e_{\mathbf{k}}, u}$. Note that the vectors $\mathbf{y}^{e_{\mathbf{k}}, u}$ are identical for all $\mathbf{k}$ whose u^{th} components coincide.

The following orthogonality property of the high dimensional Fourier coefficients is a simple corollary of its one dimensional version in [3].

Corollary 1 (Fourier orthogonality). For any $\mathbf{k} \neq \mathbf{k}' \in \mathbb{Z}^d$, we have $\mathbb{E}_s[\widehat{f}_s(\mathbf{k})^* \widehat{f}_s(\mathbf{k}')] = 0$.

Now we can compute the ℓ_2 norm as follows.

$$\begin{aligned} \mathbb{E}_s \left[\left\| (\overline{P}_h^\top)^{\otimes u} \cdot \mathbf{y}^{f_s, u} \right\|_2^2 \right] &= \mathbb{E}_s \left[\left\| (\overline{P}_h^\top)^{\otimes u} \cdot \left(\sum_{\mathbf{k} \neq \mathbf{0}} \widehat{f}_s(\mathbf{k}) \mathbf{y}^{e_{\mathbf{k}}, u} \right) \right\|_2^2 \right] \\ &= \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \cdot \prod_{j \in u} |\overline{P}_h^\top \mathbf{y}^{e_{\mathbf{k}}, j}|^2 \leq \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \cdot \prod_{j \in u} |k_j|, \end{aligned}$$

where the expectation over $\mathbf{s}$ is removed in the second line since the modulus of the Fourier coefficients do not depend on the random shift, and the last step follows from the estimate in [3, Lemma 5.3].

Now summing over all $\emptyset \neq u \subseteq [d]$ gives the following ℓ_2 bound

$$\begin{aligned} \mathbb{E}_s \left[\sum_{\emptyset \neq u \subseteq [d]} \frac{1}{\gamma(u)^2} \left\| (\overline{P}_h^\top)^{\otimes u} \mathbf{y}^{f_s, u} \right\|_2^2 \right] &\leq \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \frac{1}{\gamma(u)^2} \prod_{j \in u} |k_j| \right) \\ &= \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} \frac{|k_j|}{\gamma_j^2} \right) \\ &=: \sigma_{\text{WSO}}(f)^2. \end{aligned} \tag{A.8}$$

Using the ℓ_2 -norm bound (A.8) in (A.6) and (A.7), we obtain the bound

$$\mathbb{E} \left[\left(\sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_s) \right)^2 \right] \leq O \left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n^2} \cdot \sigma_{\text{WSO}}(f)^2 \right). \tag{A.9}$$

Plugging (A.9) and (A.4) into (A.2), we obtain that

$$\begin{aligned} \mathbb{E}[(\text{err}(A_T, f_s))^2] &\leq \mathbb{E}[(\text{err}(A_0, f_s))^2] + \mathbb{E} \left[\left(\sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_s) \right)^2 \right] \\ &\leq \frac{\sigma(f)^2}{n^2} + O \left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n^2} \cdot \sigma_{\text{WSO}}(f)^2 \right) \\ &= O \left(\frac{d \log(dn) \prod_{j \in [d]} (1 + \gamma_j^2 \log n)}{n^2} \cdot \sigma_{\text{WSO}}(f)^2 \right). \end{aligned}$$

This is the error guaranteed by the algorithm in [3], applied for the weighted discrepancy. This completes the proof of the theorem. $\square$

A.2. Proofs of results in Section 3

Proof of Theorem 2. Let $\mathcal{F}_s$ be the class of functions involving interactions of order at most s , i.e., $\mathcal{F}_s$ consists functions of the form $\sum_{|u| \leq s} f_u(\mathbf{z}^u)$. Fix an arbitrary 1-periodic function

$$f(\mathbf{z}) = \sum_{|u| \leq s} f_u(\mathbf{z}^u) \in \mathcal{F}_s$$

with a continuous mixed derivative. We can equivalently view WSubgTrans as fixing $\mathcal{D}_{\leq h}^{\otimes d}$ and applying a random shift $\mathbf{s} \sim [0, 1]^d$ to f which results in $f_s(\mathbf{z}) = f(\mathbf{z} + \mathbf{s}) = \sum_{|u| \leq s} f_u(\mathbf{z}^u + \mathbf{s}^u)$. Denote $\mathbb{E}_s$ the expectation w.r.t. random shift $\mathbf{s}$. By Hlawka-Zaremba formula, the integration error using point set A_T is

$$\begin{aligned} \text{err}(A_T, f_s) &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u f_s(\mathbf{z}) d\mathbf{z} \\ &= \sum_{\emptyset \neq u \subseteq [d]} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u \left(\sum_{|u'| \leq s} f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) \right) d\mathbf{z} \\ &= \sum_{0 < |u| \leq s} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u \left(\sum_{u': u \subseteq u'} f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) \right) d\mathbf{z}. \end{aligned}$$

By Lemma 4, we can write

$$\begin{aligned} \text{err}(A_T, f_s) &= \sum_{0 < |u| \leq s} (-1)^{|u|-1} \int_{Q_u} h_T(\mathbf{z}) \cdot \partial_u \left(\sum_{u': u \subseteq u'} f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) \right) d\mathbf{z} \\ &= \sum_{0 < |u| \leq s} (-1)^{|u|-1} \int_{Q_u} h_0(\mathbf{z}) \cdot \partial_u \left(\sum_{u': u \subseteq u'} f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) \right) d\mathbf{z} \\ &\quad + \sum_{t=0}^{T-1} \left(\sum_{0 < |u| \leq s} (-1)^{|u|-1} \int_{Q_u} \frac{\text{disc}_t(\mathbf{z})}{n_t} \cdot \partial_u \left(\sum_{u': u \subseteq u'} f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) \right) d\mathbf{z} \right) \\ &=: \text{err}(A_0, f_s) + \sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_s). \end{aligned} \tag{A.10}$$

First, since A_0 consists of n^2 independent random samples,

$$\mathbb{E}[(\text{err}(A_0, f_s))^2] \leq \frac{\sigma(f)^2}{n^2}. \tag{A.11}$$

We next analyze the discrepancy error $\text{err}_t^{\text{disc}}$ in (A.10). Using the same notation and argument as in the proof of Theorem 1 in Appendix A.1, we further write

$$\begin{aligned} \text{err}_t^{\text{disc}}(f_s) &= \sum_{0 < |u| \leq s} \sum_{u': u \subseteq u'} (-1)^{|u|-1} \int_{Q_u} \frac{\text{disc}_t(\mathbf{z})}{n_t} \cdot \partial_u f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'}) d\mathbf{z} \\ &= \frac{1}{n_t} \sum_{0 < |u| \leq s} \sum_{u': u \subseteq u'} (-1)^{|u|-1} \cdot \langle \mathbf{d}^{t, \mathcal{C}, u}, \mathbf{y}^{f_{u'}, s, u} \rangle \\ &= \frac{1}{n_t} \sum_{0 < |u| \leq s} \sum_{u': u \subseteq u'} (-1)^{|u|-1} \cdot \langle \mathbf{d}^{t, \mathcal{D}, u}, (P_h^\top)^{\otimes u} \mathbf{y}^{f_{u'}, s, u} \rangle \end{aligned}$$

$$= \frac{1}{n_t} \sum_{0 < |u| \leq s} (-1)^{|u|-1} \cdot \langle \mathbf{d}^{t, \mathcal{D}, u}, (P_h^\top)^{\otimes u} \left(\sum_{u': u \subseteq u'} \mathbf{y}^{f_{u', s}, u} \right) \rangle, \quad (\text{A.12})$$

where $f_{u', s}(\mathbf{z}) := f_{u'}(\mathbf{z}^{u'} + \mathbf{s}^{u'})$, and $\mathbf{d}^{t, \mathcal{D}, u} \in \mathbb{Z}^{|\mathcal{D}_{\leq h} \setminus C_1|^{\otimes u}|}$ is the restriction of $\mathbf{d}^{t, \mathcal{D}}$ to dyadic boxes in Q_u excluding those with any dimension in u being $(0, 1]$. It follows that the vectors $\mathbf{d}^{t, \mathcal{D}, u}$ for different u with $0 < |u| \leq s$ correspond to disjoint coordinates of the vector $\mathbf{d}^{t, \mathcal{D}}|_{\tilde{\mathcal{D}}_{\leq h}^{\otimes s}}$, which is the restriction of $\mathbf{d}^{t, \mathcal{D}}$ to $\tilde{\mathcal{D}}_{\leq h}^{\otimes s}$.

Recall that in our modified WSubgTrans algorithm, the combinatorial discrepancy vector is defined with respect to $\tilde{\mathcal{D}}_{\leq h}^{\otimes s}$ rather than the full collection $\mathcal{D}_{\leq h}^{\otimes d}$. Moreover,

$$\begin{aligned} \|\mathbf{v}_j\|_2^2 &= 1 + \sum_{B \in \tilde{\mathcal{D}}_{\leq h}^{\otimes s}} \mathbf{1}_{\{z_j \in B\}} = 1 + \sum_{u \in [d]: |u| \leq s} h^{|u|} = 1 + \sum_{k=0}^s \binom{d}{k} h^k = O_s(d^s h^s) \\ &= O_s((d \log(dn))^s). \end{aligned}$$

Analogous to Lemma 5, we conclude that for any $0 \leq t \leq T-1$, the random vector $(\mathbf{c}_t, \mathbf{d}^{t, \mathcal{D}}|_{\tilde{\mathcal{D}}_{\leq h}^{\otimes s}})$ is $O_s(d^s (\log(dn))^{s+1})$ -subgaussian.

Consequently, for any outcome of the random shift $\mathbf{s}$, $\text{err}_t^{\text{disc}}(f_s)$ in (A.12) is also subgaussian with parameter

$$\begin{aligned} &O_s\left(\frac{d^s (\log(dn))^{s+1}}{n_t^2} \cdot \sum_{0 < |u| \leq s} \|(P_h^\top)^{\otimes u} \left(\sum_{u': u \subseteq u'} \mathbf{y}^{f_{u', s}, u} \right)\|_2^2\right) \\ &\leq O_s\left(\frac{d^s (\log(dn))^{s+1}}{n_t^2} \cdot \sum_{0 < |u| \leq s} \|(\bar{P}_h^\top)^{\otimes u} \left(\sum_{u': u \subseteq u'} \mathbf{y}^{f_{u', s}, u} \right)\|_2^2\right). \end{aligned} \quad (\text{A.13})$$

For a fixed subset $u \subseteq [d]$ with $0 < |u| \leq s$, we have

$$\begin{aligned} \mathbb{E}_{\mathbf{s}} \left[\left\| (\bar{P}_h^\top)^{\otimes u} \left(\sum_{u': u \subseteq u'} \mathbf{y}^{f_{u', s}, u} \right) \right\|_2^2 \right] &= \mathbb{E}_{\mathbf{s}} \left[\left\| (\bar{P}_h^\top)^{\otimes u} \cdot \left(\sum_{u': u \subseteq u'} \sum_{\mathbf{k} \neq \mathbf{0}} \widehat{f_{u', s}}(\mathbf{k}) \mathbf{y}^{e_{\mathbf{k}}, u} \right) \right\|_2^2 \right] \\ &= \mathbb{E}_{\mathbf{s}} \left[\left\| (\bar{P}_h^\top)^{\otimes u} \cdot \left(\sum_{\mathbf{k} \neq \mathbf{0}} \left(\sum_{u': u \subseteq u'} \widehat{f_{u', s}}(\mathbf{k}) \right) \mathbf{y}^{e_{\mathbf{k}}, u} \right) \right\|_2^2 \right] \\ &= \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': u \subseteq u'} \widehat{f_{u', s}}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |\bar{P}_h^\top \mathbf{y}^{e_{\mathbf{k}}, j}|^2 \\ &\leq \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': u \subseteq u'} \widehat{f_{u', s}}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j|, \end{aligned}$$

where we have used the Fourier orthogonality (Corollary 1) and the estimate [3, Lemma 5.3], and the expectation over $\mathbf{s}$ is removed in the third line since the modulus of the Fourier coefficients do not depend on the random shift.

Now summing over all subsets $u \subseteq [d]$ satisfying $0 < |u| \leq s$ gives the following ℓ_2 bound

$$\mathbb{E}_{\mathbf{s}} \left[\sum_{0 < |u| \leq s} \left\| (\bar{P}_h^\top)^{\otimes u} \left(\sum_{u': u \subseteq u'} \mathbf{y}^{f_{u', s}, u} \right) \right\|_2^2 \right] \leq \sum_{0 < |u| \leq s} \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': u \subseteq u'} \widehat{f_{u', s}}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j|. \quad (\text{A.14})$$

For $\mathbf{k} = (k_1, \dots, k_d) \in \mathbb{Z}^d$, we denote its support by $\text{supp}(\mathbf{k}) := \{j \in [d] : k_j \neq 0\}$. By the definition of Fourier transform, $\widehat{f_{u', s}}(\mathbf{k}) = 0$ if $\text{supp}(\mathbf{k})$ is not a subset of u' . Therefore,

$$\sum_{0 < |u| \leq s} \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': u \subseteq u'} \widehat{f_{u', s}}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j|$$

$$\begin{aligned}
&= \sum_{\mathbf{k} \neq \mathbf{0}} \sum_{0 < |u| \leq s} \left| \sum_{u': u \subseteq u'} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j| \\
&= \sum_{\mathbf{k} \neq \mathbf{0}} \sum_{0 < |u| \leq s, u \subseteq \text{supp}(\mathbf{k})} \left| \sum_{u': u \subseteq u', \text{supp}(\mathbf{k}) \subseteq u'} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j| \\
&= \sum_{\mathbf{k} \neq \mathbf{0}} \sum_{0 < |u| \leq s, u \subseteq \text{supp}(\mathbf{k})} \left| \sum_{u': \text{supp}(\mathbf{k}) \subseteq u'} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \cdot \prod_{j \in u} |k_j| \\
&= \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': \text{supp}(\mathbf{k}) \subseteq u'} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \left(\sum_{0 < |u| \leq s, u \subseteq \text{supp}(\mathbf{k})} \prod_{j \in u} |k_j| \right) \\
&\stackrel{(\star)}{=} \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{u': \text{supp}(\mathbf{k}) \subseteq u'} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} |k_j| \right) \\
&= \sum_{\mathbf{k} \neq \mathbf{0}} \left| \sum_{|u'| \leq s} \widehat{f}_{u'}(\mathbf{k}) \right|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} |k_j| \right) \\
&= \sum_{\mathbf{k} \neq \mathbf{0}} |\widehat{f}(\mathbf{k})|^2 \left(\sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} |k_j| \right) \\
&=: \sigma_{\text{SO}}(f)^2,
\end{aligned} \tag{A.15}$$

where $(\star)$ follows because the sum over $\mathbf{k} \neq \mathbf{0}$ effectively restricts to frequencies with $|\text{supp}(\mathbf{k})| \leq s$. Hence,

$$\sum_{0 < |u| \leq s, u \subseteq \text{supp}(\mathbf{k})} \prod_{j \in u} |k_j| = \sum_{\emptyset \neq u \subseteq \text{supp}(\mathbf{k})} \prod_{j \in u} |k_j| = \sum_{\emptyset \neq u \subseteq [d]} \prod_{j \in u} |k_j|.$$

Using the ℓ_2 -norm bound (A.14) and the equality (A.15) in (A.12) and (A.13), we obtain the bound

$$\mathbb{E} \left[\left(\sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_s) \right)^2 \right] \leq O_s \left(\frac{d^s (\log(dn))^{s+1}}{n^2} \cdot \sigma_{\text{SO}}(f)^2 \right). \tag{A.16}$$

Plugging (A.16) and (A.11) into (A.10), we obtain that for any $f(\mathbf{z}) = \sum_{|u| \leq s} f_u(\mathbf{z}^u) \in \mathcal{F}_s$,

$$\begin{aligned}
\mathbb{E}[(\text{err}(A_T, f_s))^2] &\leq \mathbb{E}[(\text{err}(A_0, f_s))^2] + \mathbb{E} \left[\left(\sum_{t=0}^{T-1} \text{err}_t^{\text{disc}}(f_s) \right)^2 \right] \\
&\leq \frac{\sigma(f)^2}{n^2} + O_s \left(\frac{d^s (\log(dn))^{s+1}}{n^2} \cdot \sigma_{\text{SO}}(f)^2 \right) \\
&= O_s \left(\frac{d^s (\log(dn))^{s+1}}{n^2} \cdot \sigma_{\text{SO}}(f)^2 \right).
\end{aligned} \tag{A.17}$$

A slight modification of [3, Lemma 4.3] yields that, for any $f : [0, 1]^d \rightarrow \mathbb{R}$,

$$\mathbb{E}[(\text{err}(A_T, f))^2] \leq O \left(\frac{\log n}{n} \sigma(f)^2 \right). \tag{A.18}$$

For any decomposition $f = g + h$ with $g, h : [0, 1]^d \rightarrow \mathbb{R}$ and $h \in \mathcal{F}_s$, we have $\text{err}(A_T, f) = \text{err}(A_T, g) + \text{err}(A_T, h)$. The theorem then follows directly from (A.17) and (A.18). $\square$

Proof of Theorem 3. In this case, $\gamma_1 = \dots = \gamma_s = 1, \gamma_{s+1} = \dots = \gamma_d = 0$, and thus

$$\|\mathbf{v}_j\|_2^2 = 1 + \prod_{j \in [d]} (1 + h\gamma_j^2) = O(h^s) = O((\log(dn))^s).$$

Analogous to Lemma 5, we conclude that for any $0 \leq t \leq T - 1$, the random vector $(\mathbf{c}_t, \mathbf{w}^{t,D})$ is $O_s((\log(dn))^{s+1})$ -subgaussian. The result follows from a slight modification of the arguments in the proofs of Theorems 1 and 2; we omit the proof for brevity. $\square$

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